

Module 3_Sequential circuits

3.3_Shift registers

Introduction: Registers

- An n -bit register has a group of n flip-flops and some logic gates and is capable of storing n bits of information.
- The flip-flops store the information while the gates control when and how new information is transferred into the register.
- Some functions of register:
 - ❖ retrieve data from register
 - ❖ store/load new data into register (serial or parallel)
 - ❖ shift the data within register (left or right)

Shift Registers

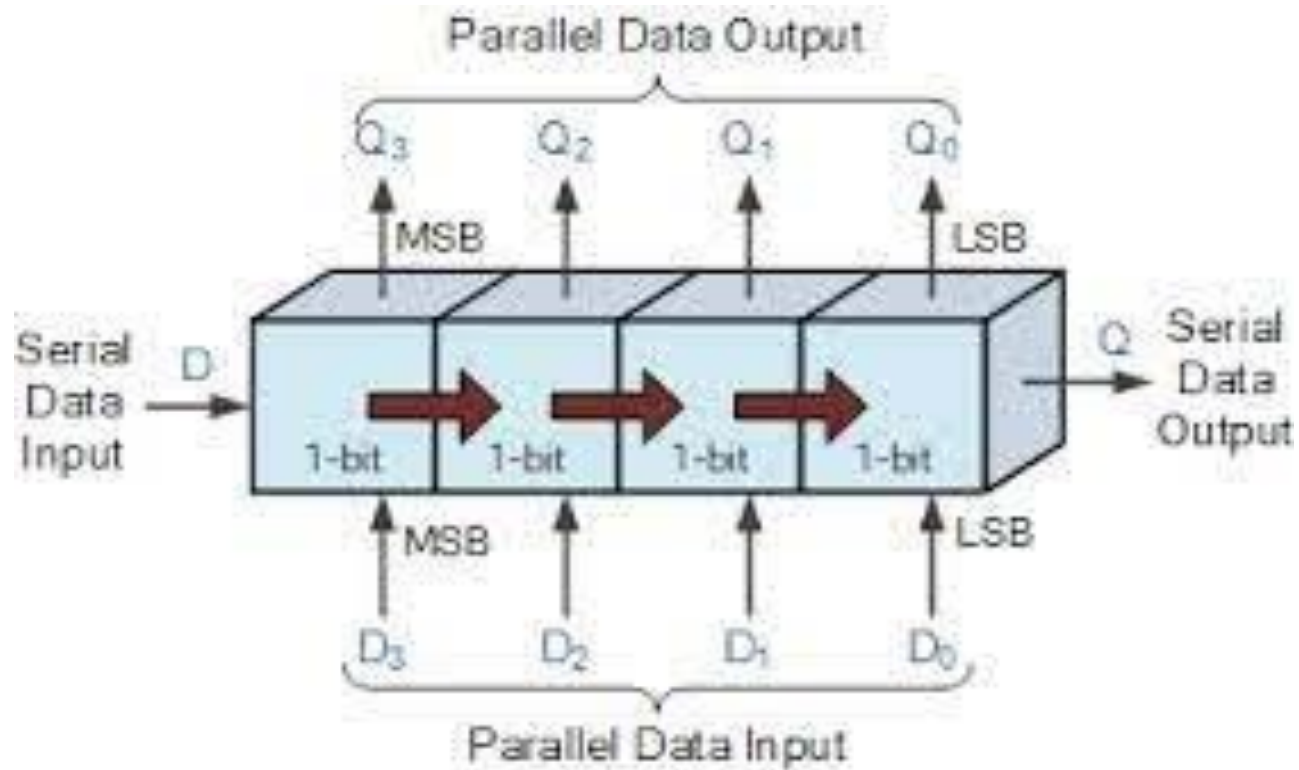
- The information stored within these registers can be transferred with the help of **shift registers**.
- Shift Register is a group of flip flops used to store multiple bits of data.
- The bits stored in such registers can be made to move within the registers and in/out of the registers by applying clock pulses.
- An n-bit shift register can be formed by connecting n flip-flops where each flip flop stores a single bit of data.
- The registers which will shift the bits to left are called “Shift left registers”.
- The registers which will shift the bits to right are called “Shift right registers”.

Types of Shift registers

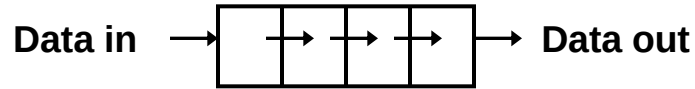
Shift registers are basically of 4 types.

- Serial In Serial Out shift register (SISO)
- Serial In parallel Out shift register(SIPO)
- Parallel In Serial Out shift register(PISO)
- Parallel In parallel Out shift register(PIPO)

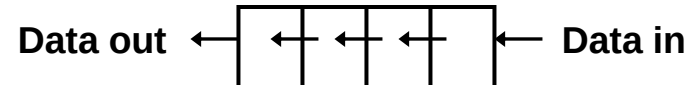
General schematic diagram of shift register



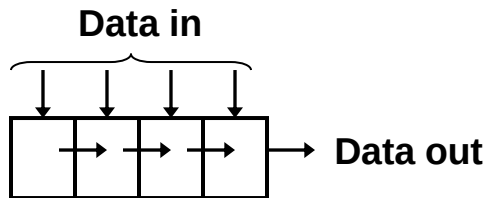
Basic data movement in shift registers (four bits are used for illustration).



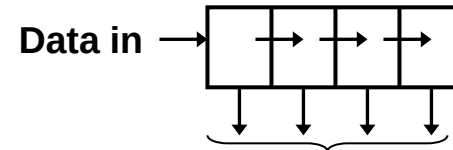
(a) Serial in/shift right/serial out



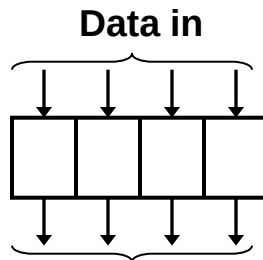
(b) Serial in/shift left/serial out



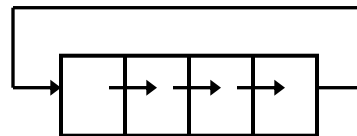
(c) Parallel in/serial out



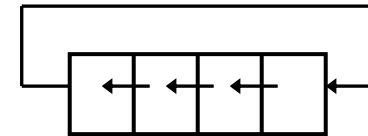
(d) Serial in/parallel out



**(e) Parallel in /
parallel out**



(f) Rotate right

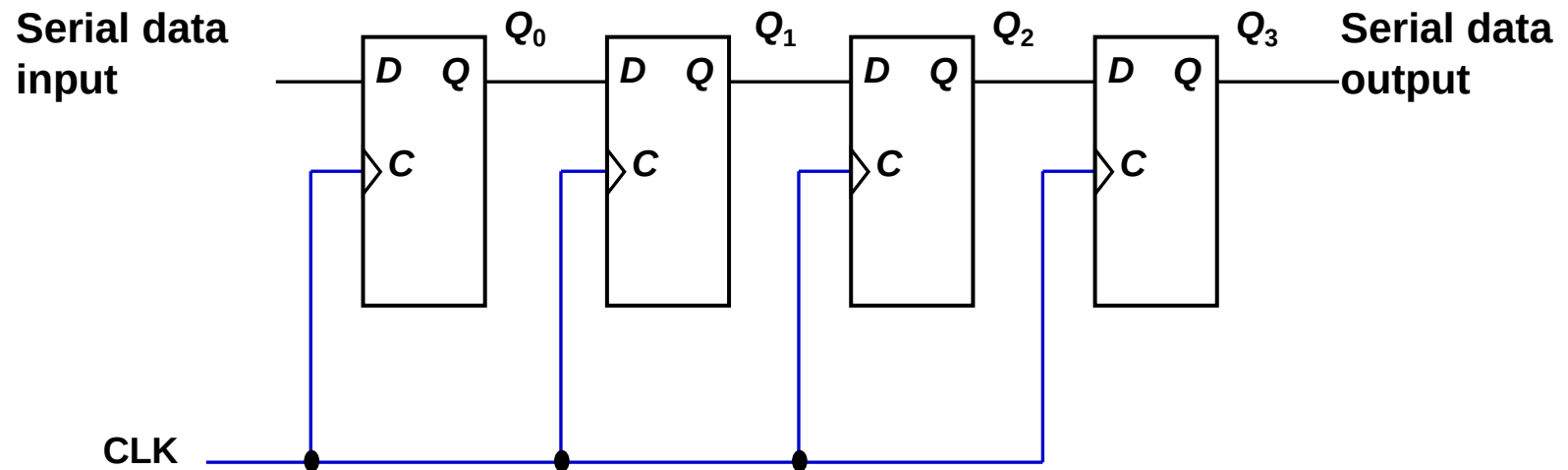


(g) Rotate left

Serial In/Serial Out Shift Registers (SISO)

- Accepts data serially — one bit at a time — and also produces output serially.
- The circuit consists of four D flip-flops which are connected in a serial manner.
- All these flip-flops are synchronous with each other since the same clock signal is applied to each flip flop

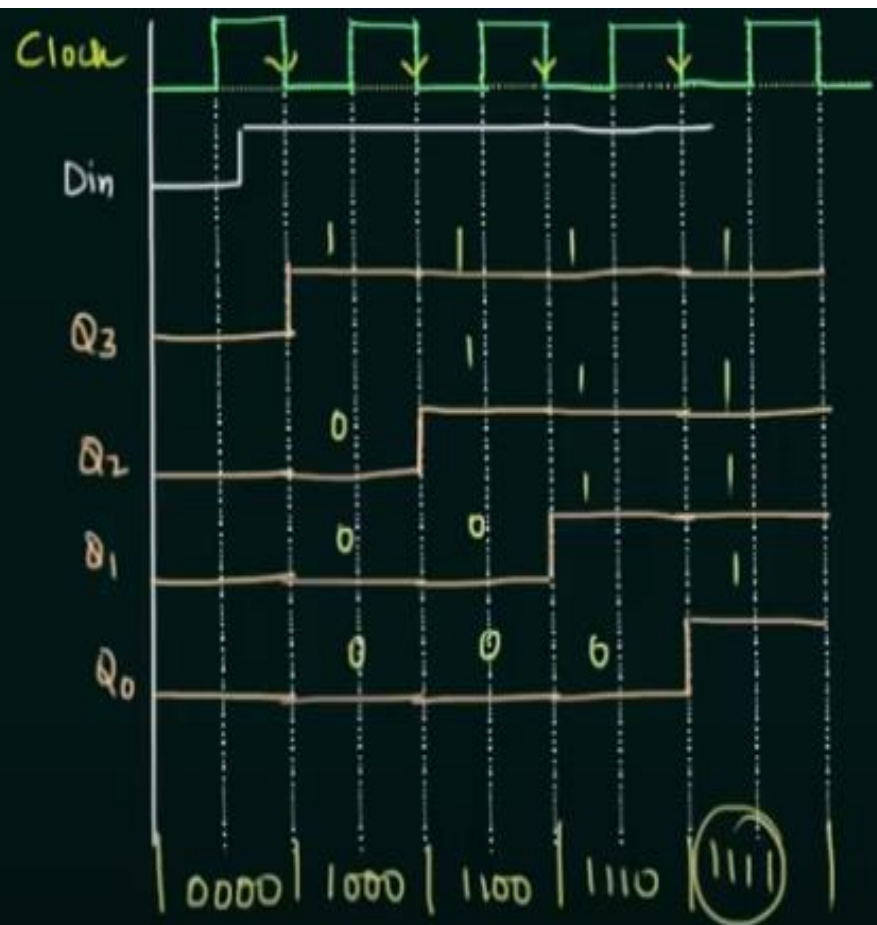
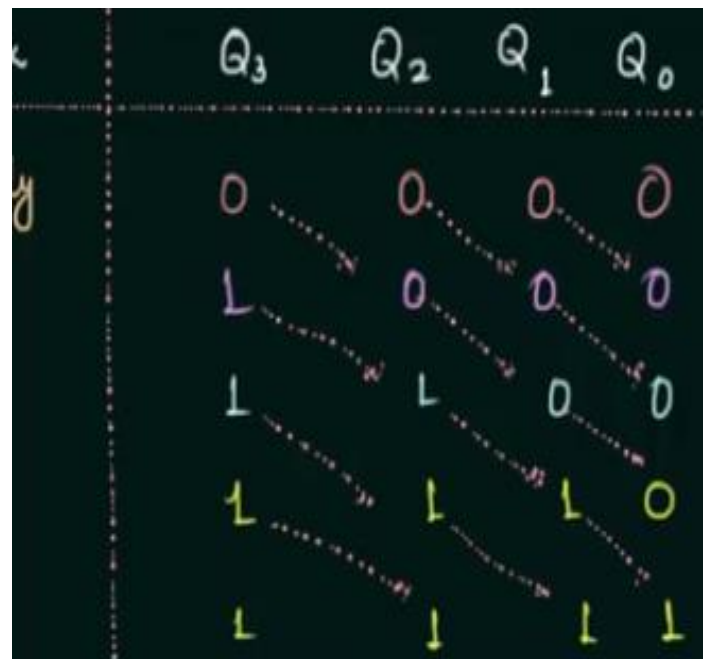
Implementation of SISO register(4 bit)



- The above circuit is an example of shift right register, taking the serial data input from the left side of the flip flop.
- The main use of a SISO is to act as a delay element.

Data storage example with “1111”

CLK	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
↓	1	0	0	0
↓	1	1	0	0
↓	1	1	1	0
↓	1	1	1	1



Application of SISO register

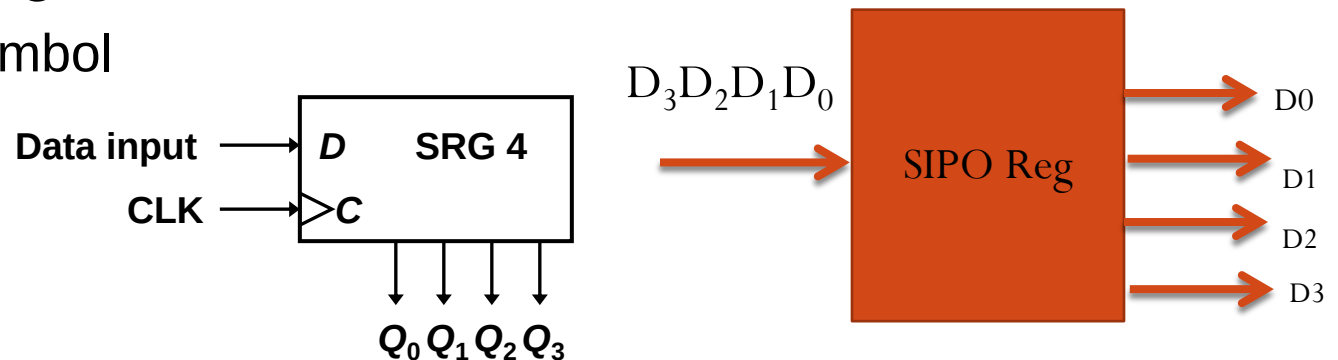
- In long distance communication for transmission of data serially from transmitter and receiver



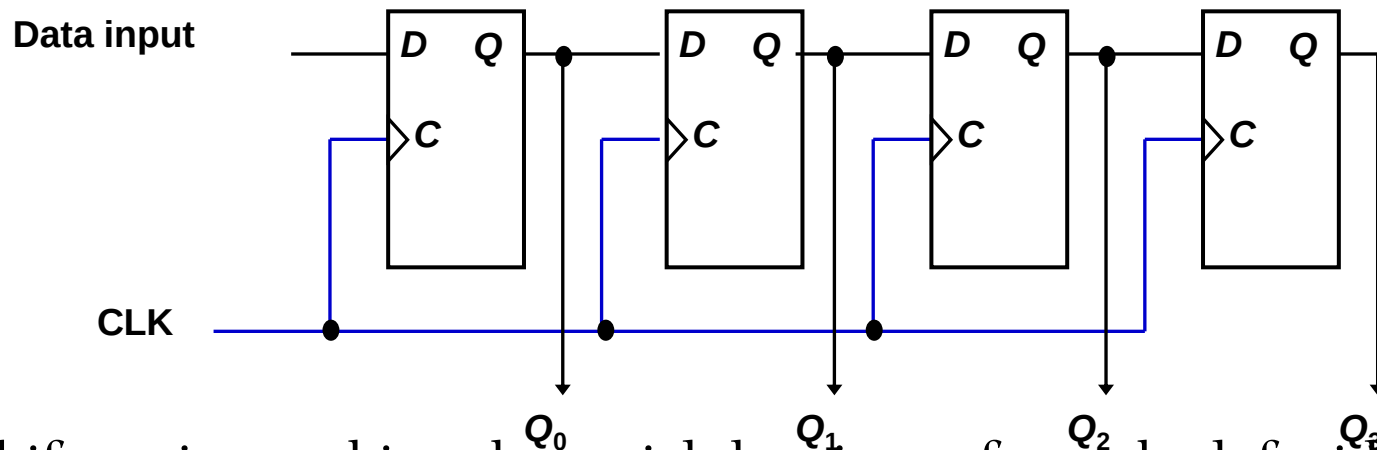
Serial In/Parallel Out Shift Registers(SIPO)

- The shift register, which allows serial input (one bit after the other through a single data line) OR Accepts data serially and produces a parallel output
- Outputs of all stages are available simultaneously.
- The output of the first flip flop is connected to the input of the next flip flop and so on.
- All these flip-flops are synchronous with each other since the same clock signal is applied to each flip flop.

Logic symbol

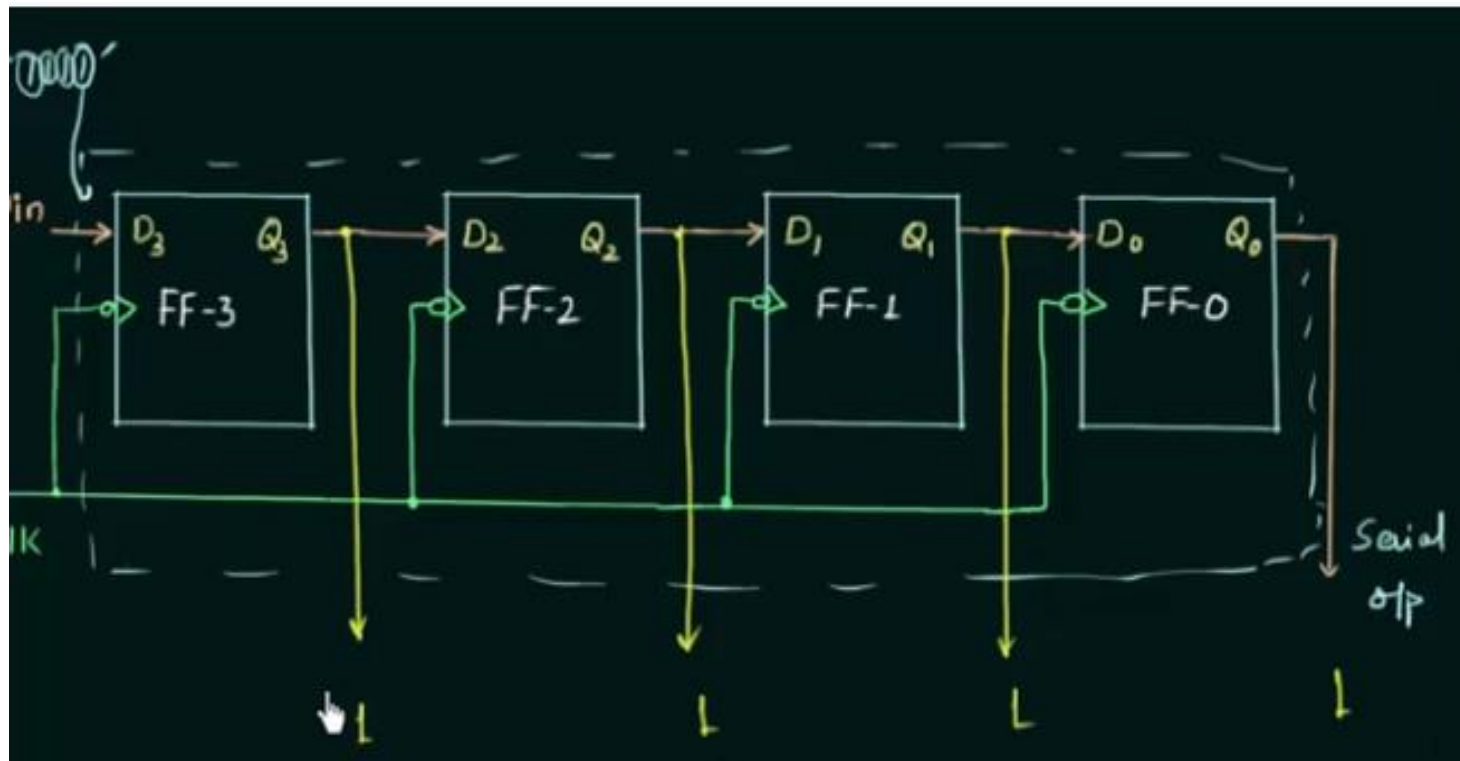


Implementation of SIPO register(4 bit)

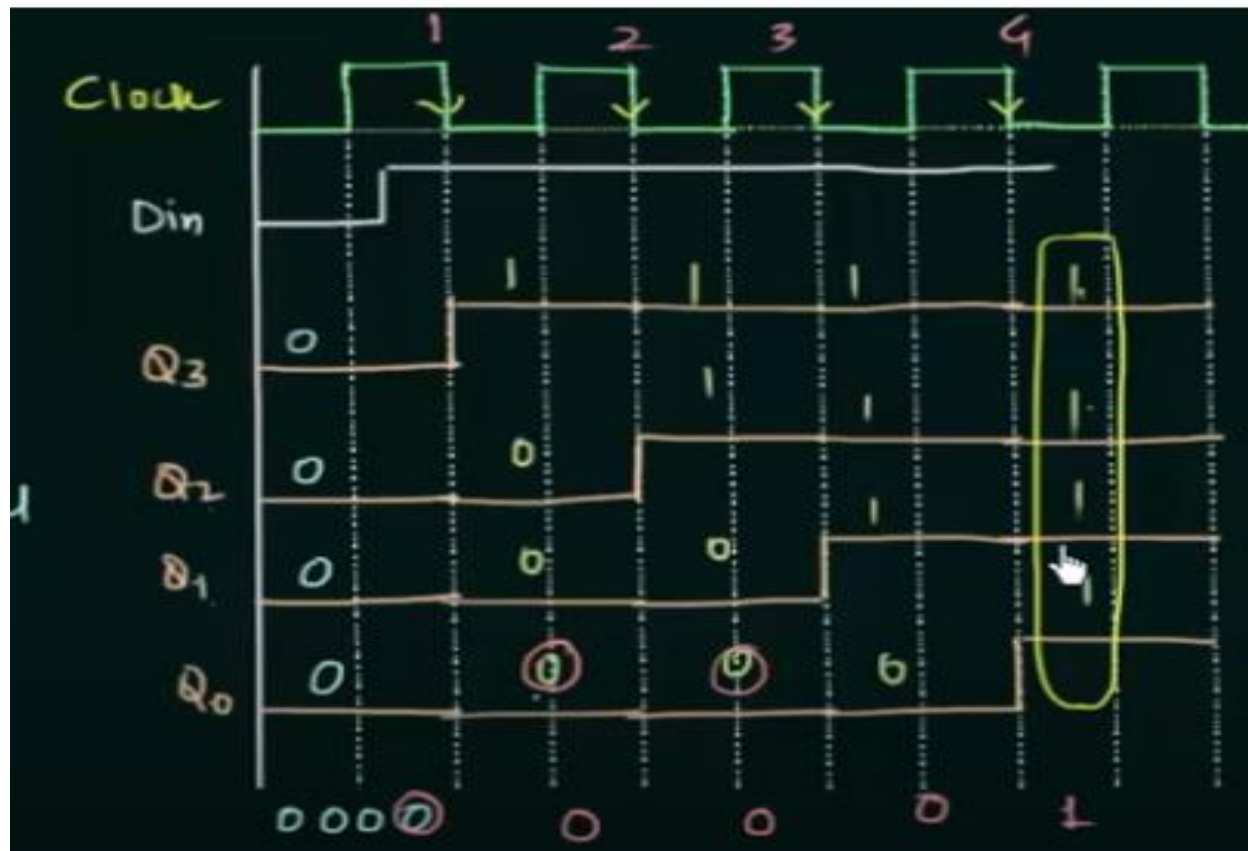


- Shift register taking the serial data input from the left side of the flip flop and producing a parallel output.
- They are used in communication lines where demultiplexing of a data line into several parallel lines is required because the main use of the SIPO register is to convert serial data into parallel data.

Data "1111" in SIPO



Timing diagram of 4 bit SISO register for data 1111



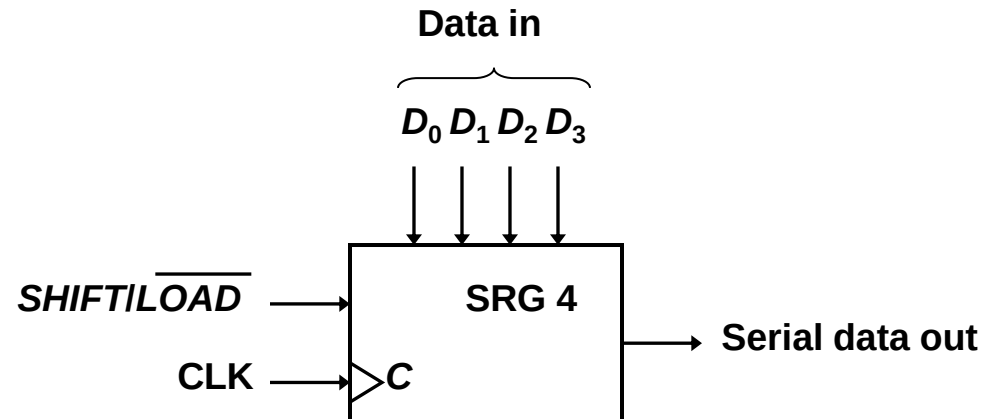
Clock pulses requirement

- In SISO and SIPO operations for n bit data storage n clock pulses are required.
- But in SISO register to take data out from SISO register more number of clock pulses are required
- In SIPO register n clock pulses are required to take data out from SIPO register

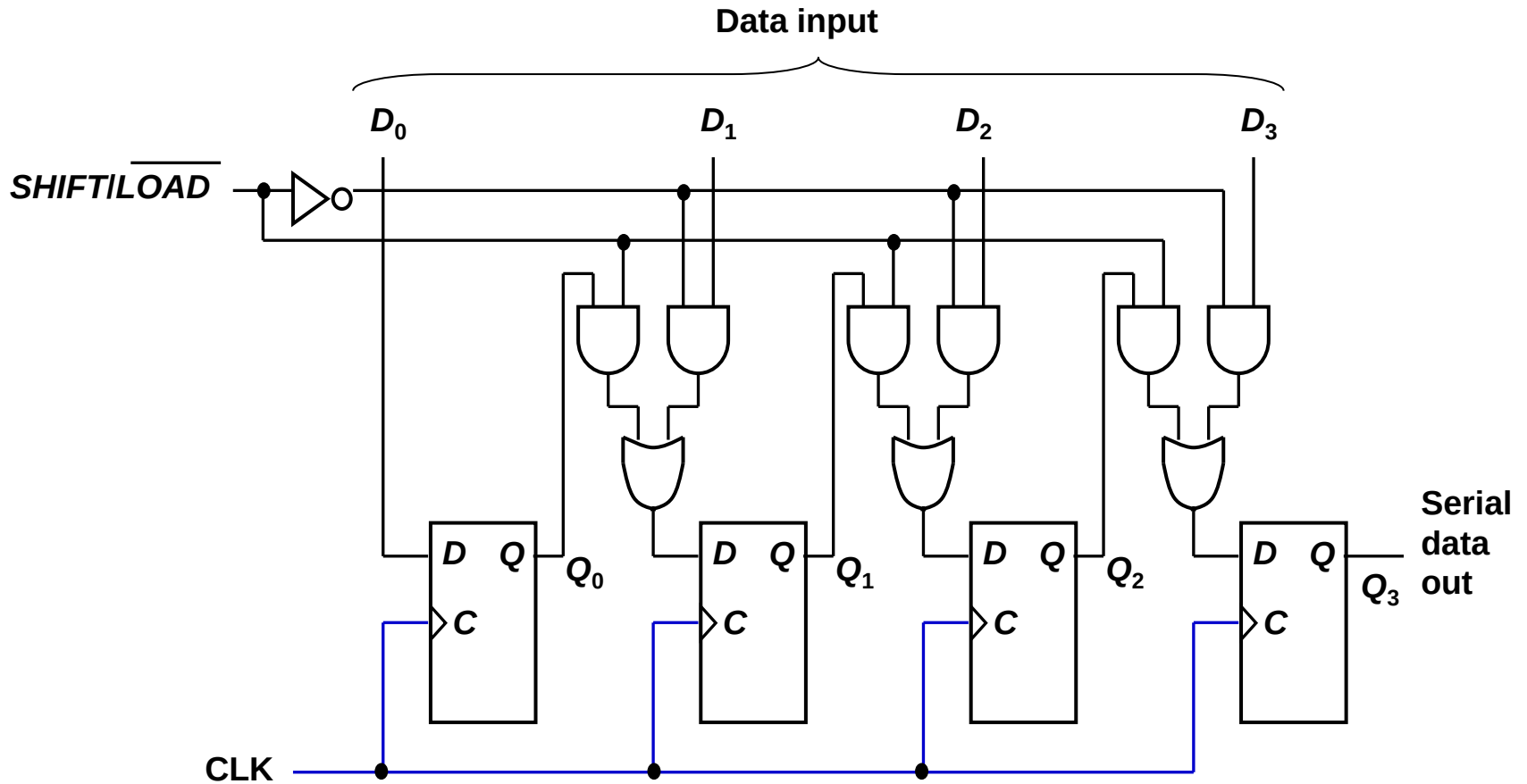
Parallel-In Serial-Out Shift Register (PISO)

- The shift register, which allows parallel input (data is given separately to each flip flop and in a simultaneous manner) and produces a serial output is known as Parallel-In Serial-Out shift register.

Logic symbol



Implementation of PISO register(4 bit)



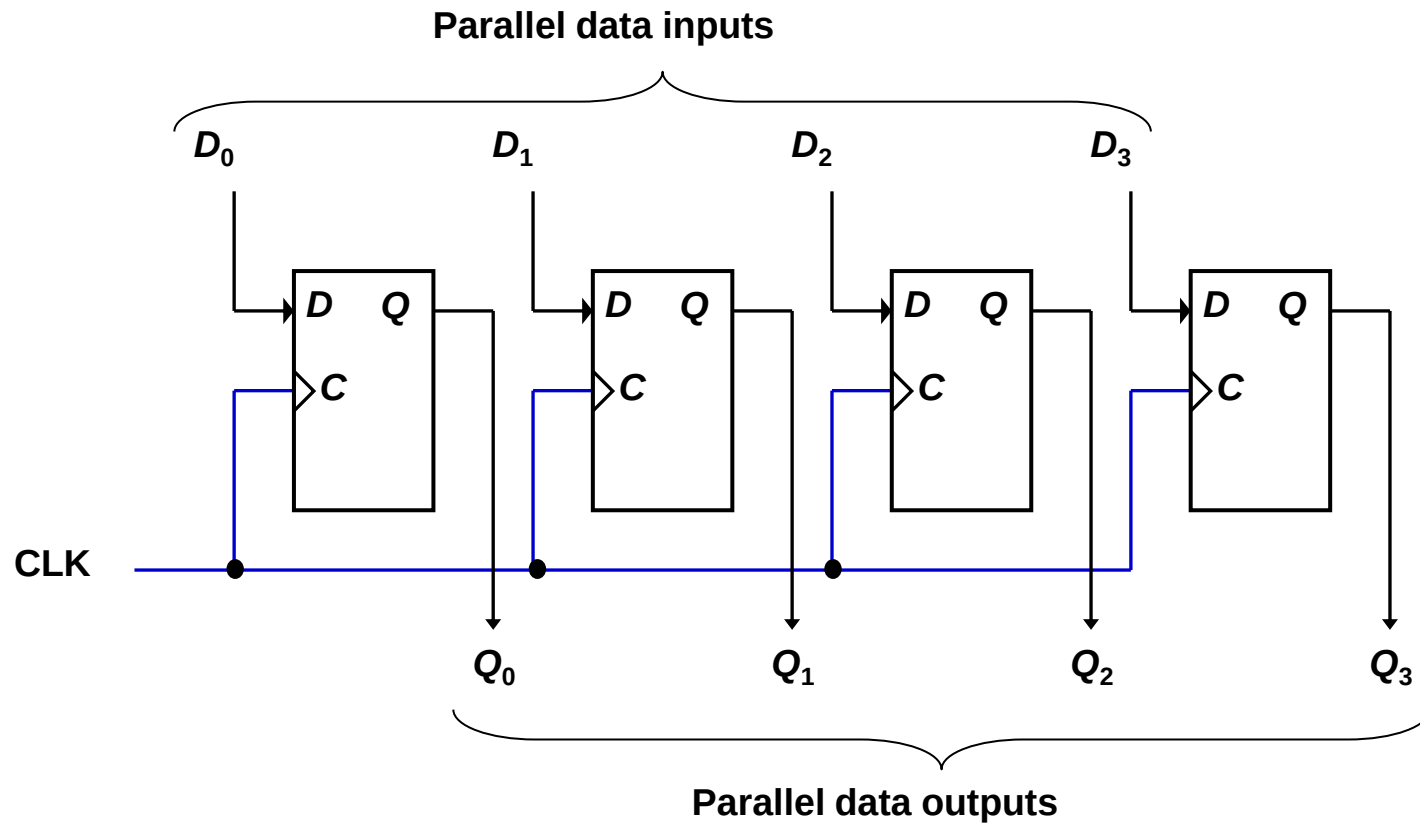
Implementation of PISO register(4 bit) cond...

- The circuit consists of four D flip-flops which are connected in series.
- . The clock input is directly connected to all the flip flops. All these flip-flops are synchronous with each other since the same clock signal is applied to each flip flop.
- The input data is connected individually to each flip flop through a multiplexer at the input of every flip flop.
- The output of the previous flip flop and parallel data input are connected to the input of the MUX and the output of MUX is connected to the next flip flop.
- A Parallel in Serial out (PISO) shift register is used to convert parallel data to serial data.

Parallel-In Parallel-Out Shift Register (PIPO) (4 bits)

- The shift register, which allows parallel input (data is given separately to each flip flop and in a simultaneous manner) and also produces a parallel output is known as Parallel-In parallel-Out shift register.
- Simultaneous input and output of all data bits
- A Parallel in Parallel out (PIPO) shift register is used as a temporary storage device and like SISO Shift register it acts as a delay element.

Implementation of PIP0 register(4 bit)



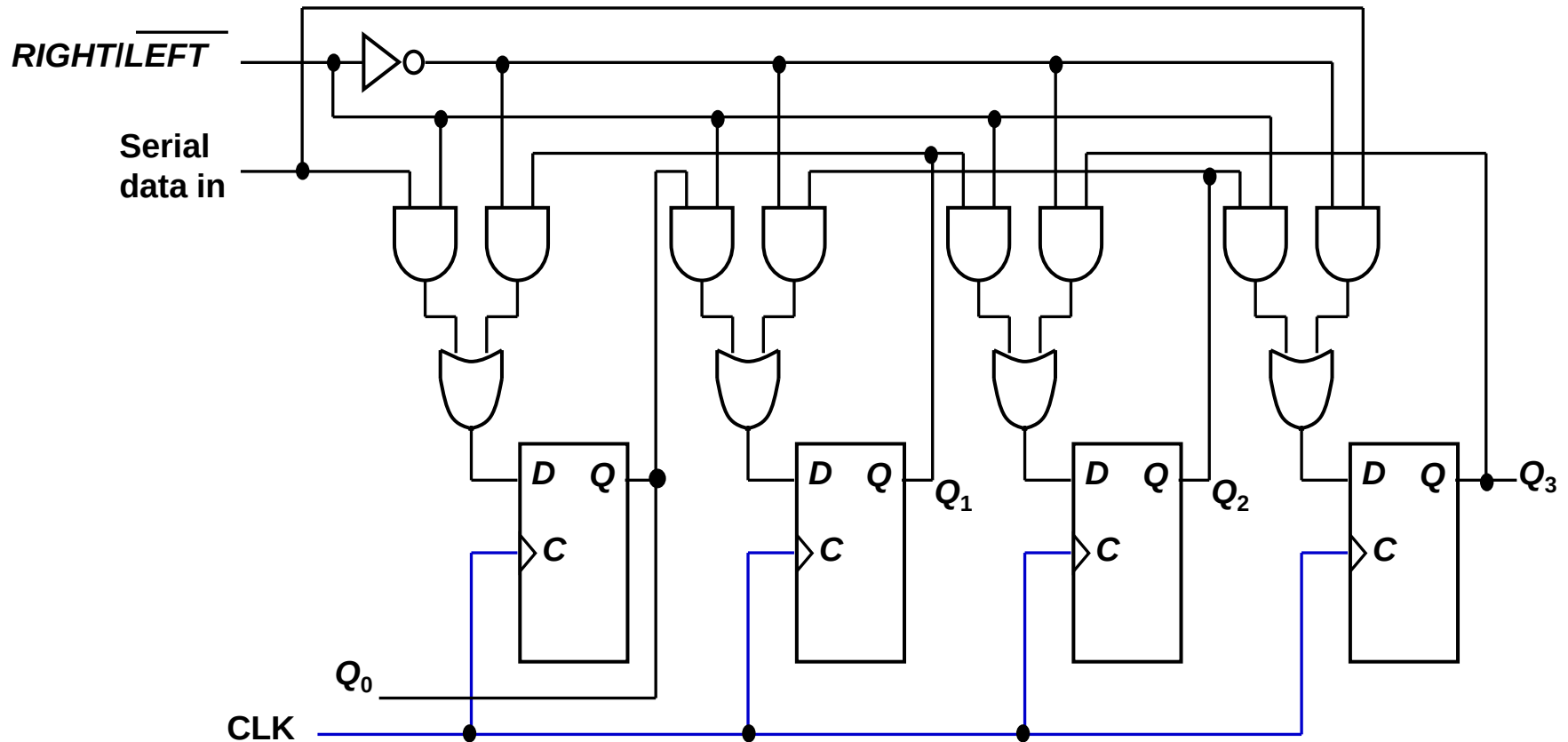
Implementation of PIP0 register(4 bit) cond...

- The circuit consists of four D flip-flops which are connected in series.
- The clear (CLR) signal and clock signals are connected to all the 4 flip flops.
- In this type of register, there are no interconnections between the individual flip-flops since no serial shifting of the data is required.
- Data is given as input separately for each flip flop and in the same way, output also collected individually from each flip flop.

Bidirectional Shift Register

- If we shift a binary number to the left by one position, it is equivalent to multiplying the number by 2 and if we shift a binary number to the right by one position, it is equivalent to dividing the number by 2. To perform these operations we need a register which can shift the data in either direction.
- Bidirectional shift registers are the registers which are capable of shifting the data either right or left depending on the mode selected. If the mode selected is 1 (high), the data will be shifted towards the right direction and if the mode selected is 0 (low), the data will be shifted towards the left direction.

Implementation of Bidirectional register(4 bit)



- The circuit consists of four D flip-flops which are connected in series
- The input data is connected at two ends of the circuit and depending on the mode selected only one and gate is in the active state

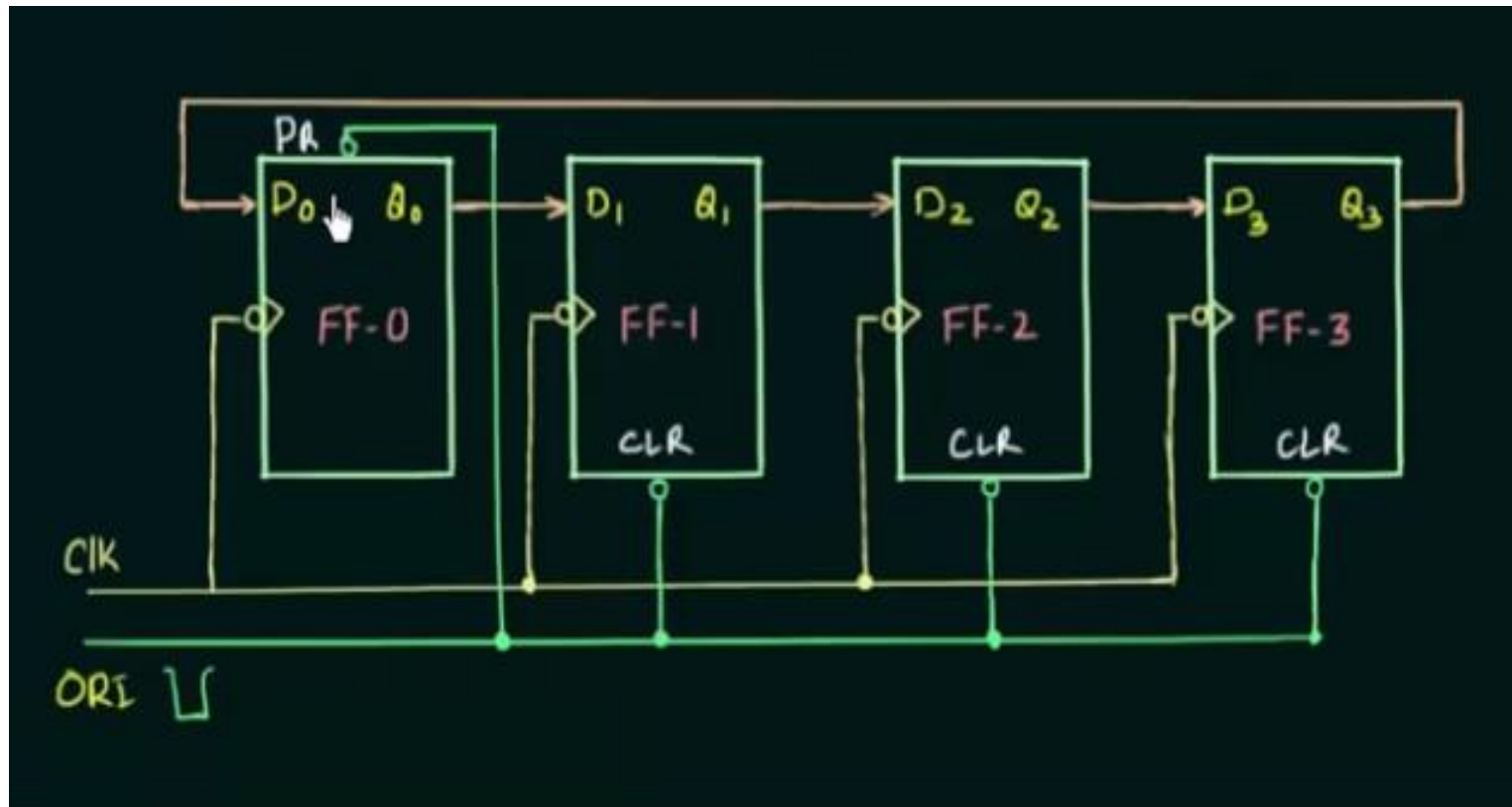
Shift Register Counters

- **Shift register counter**: a shift register with the serial output connected back to the serial input.
- They are classified as counters because they give a specified sequence of states.
- Two common types: the *Johnson counter* and the *Ring counter*.

Ring Counter (4 bit)

- The circuit consists of four D flip-flops which are connected in series.
- A Ring counter is generally used because it is self-decoding. No extra decoding circuit is needed to determine what state the counter is in.
- Ring counter is application of shift register
- Difference is :Output of last FF is connected to input of first FF
- Number of states = Number of FFs
- If n number of FFs are used in ring counter then it will count n states.

Ring counter circuit(4 states)

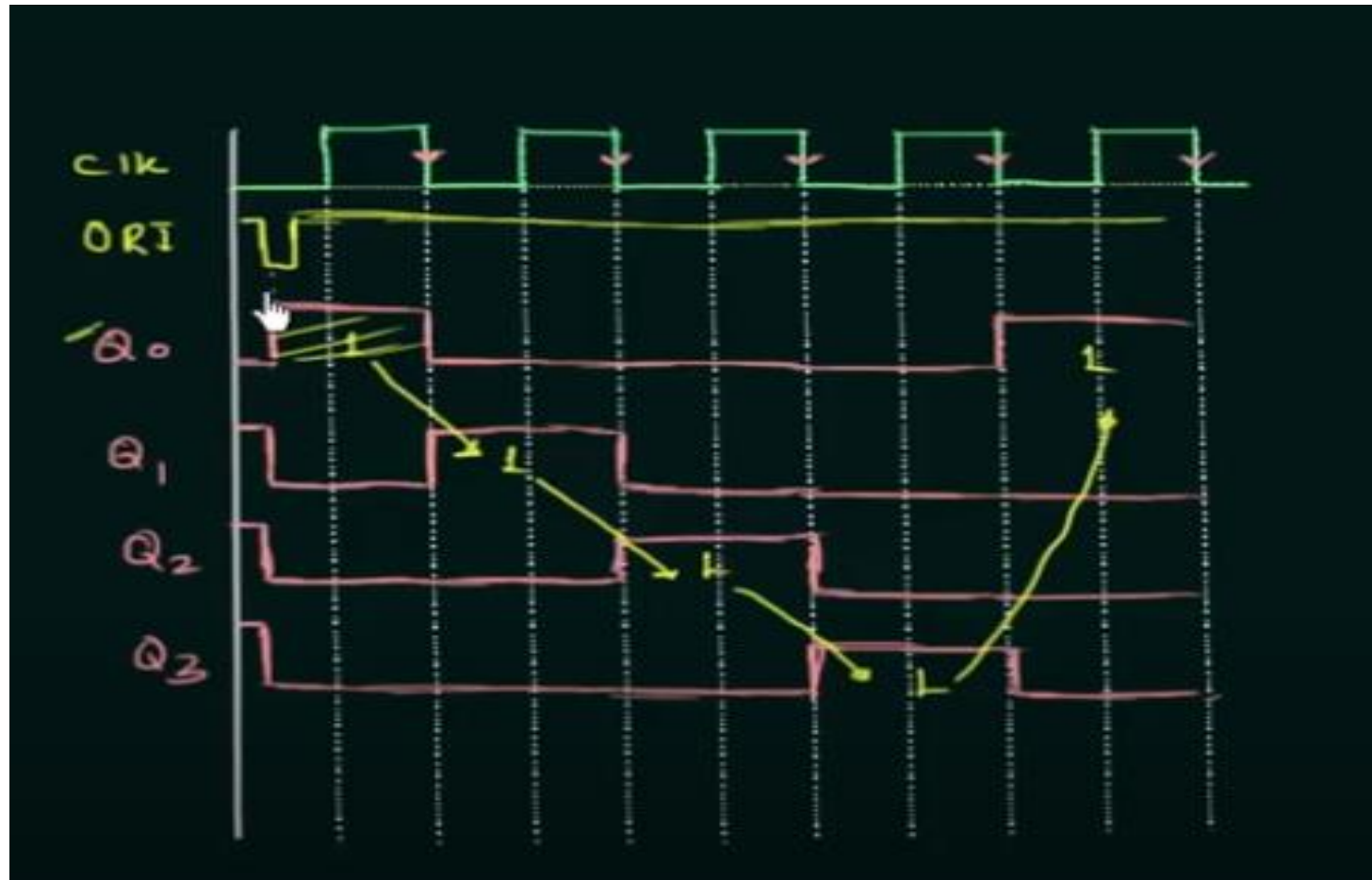


Truth table of Ring counter (4 bits)

- Since the circuit consists of four flip flops the data pattern will repeat after every four clock pulses as shown in the truth table below:

ORI	CLK	Q ₀	Q ₁	Q ₂	Q ₃
	X	1	0	0	0
1	↓	0	1	0	0
1	↓	0	0	1	0
1	↓	0	0	0	1
1	↓	1	0	0	0

Timing diagram of Ring counter (4 states)

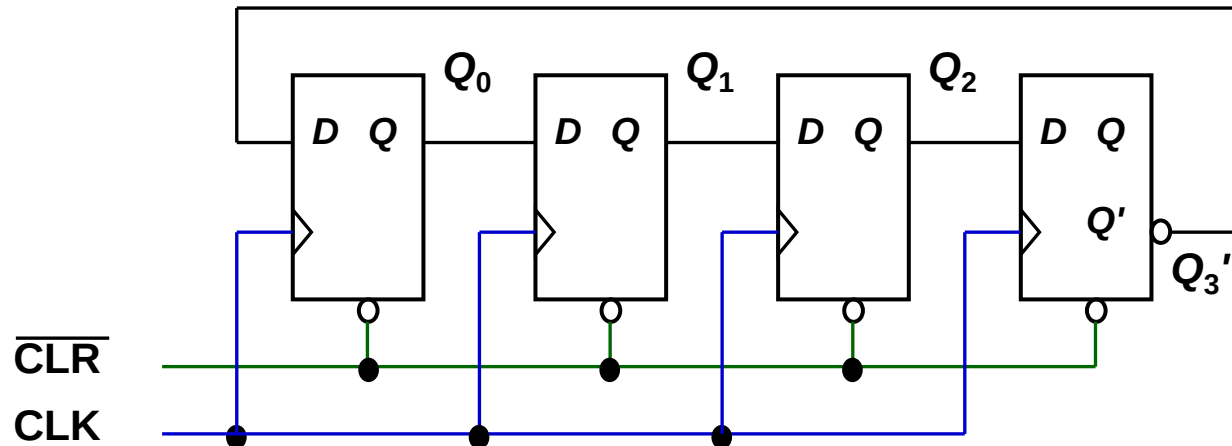


Johnson Counter (4 bits)

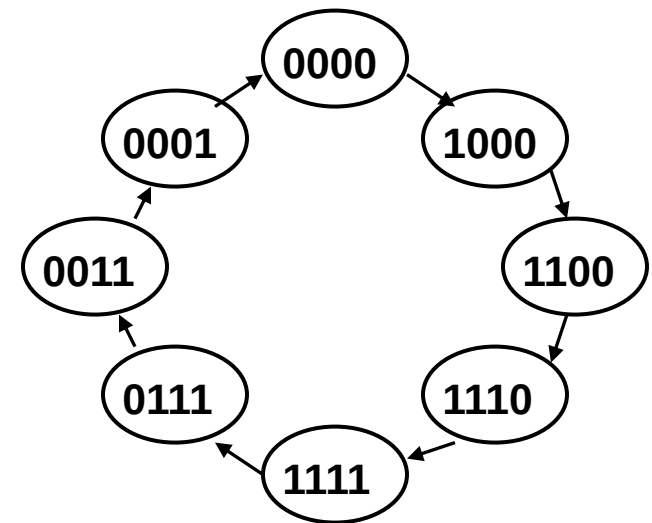
- A Johnson counter is basically a shift register counter in which the output of the first flip flop is connected to the next flip flop and so on and the inverted output of the last flip flop is again fed back to the input of the first flip flop.
- They are also known as twisted ring counters.
- The circuit consists of four D flip-flops for 4 bit counter which are connected in series.
- An n -stage Johnson counter yields a count sequence of $2n$ different states, thus also known as a mod- $2n$ counter.

- Require fewer flip-flops than ring counters but more flip-flops than binary counters.
- The main advantage of Johnson counter is that it only needs n number of flip-flops compared to the ring counter to circulate a given data to generate a sequence of $2n$ states.
- An n -bit Johnson counter cycles through $2n$ states.
- Require more decoding circuitry than ring counter but less than binary counters.

Example: A 4-bit (MOD-8) Johnson counter

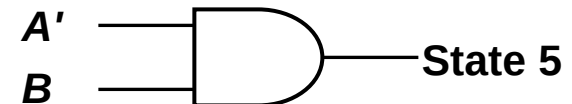
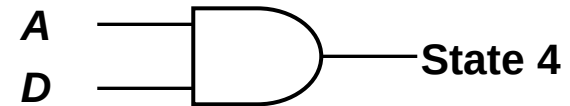
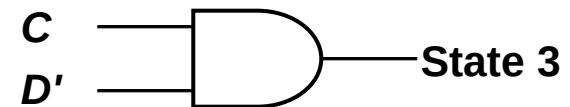
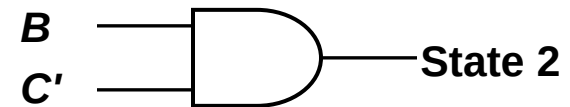
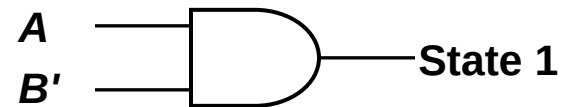
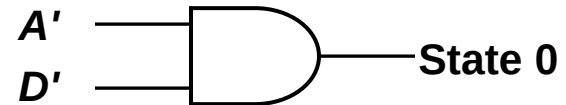
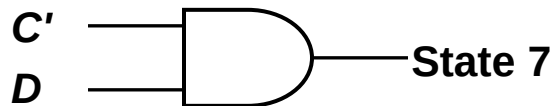
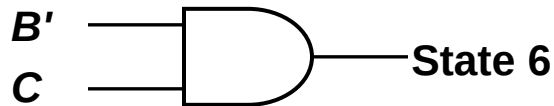


Clock	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1



Decoding logic for a 4-bit Johnson counter

Clock	A	B	C	D	Decoding
0	0	0	0	0	$A'.D'$
1	1	0	0	0	$A.B'$
2	1	1	0	0	$B.C'$
3	1	1	1	0	$C.D'$
4	1	1	1	1	$A.D$
5	0	1	1	1	$A'.B$
6	0	0	1	1	$B'.C$
7	0	0	0	1	$C'.D$



Applications of shift Registers

- The shift registers are used for temporary data storage.
- The shift registers are also used for data transfer and data manipulation.
- The serial-in serial-out and parallel-in parallel-out shift registers are used to produce time delay to digital circuits.
- The serial-in parallel-out shift register is used to convert serial data into parallel data thus they are used in communication lines where demultiplexing of a data line into several parallel line is required.
- A Parallel in Serial out shift register is used to convert parallel data to serial data.